

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

PRESTRESSED CONCRETE

(Civil Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) State difference between pre- tensioning and Post tensioning. | 2M |
| (b) What type of concrete will you prefer for the pre-stressing work system? | 2M |
| (c) Enumerate the various losses in pre-stress. | 2M |
| (d) What are the assumptions in the analysis of pre-stressed concrete? | 2M |
| (e) What is pressure line or thrust line in prestress? | 2M |
| (f) What are different stages of loading considered while designing pre-stressed concrete beam? | 2M |
| (g) What is long term deflection in beam? | 2M |
| (h) What are the factors responsible for cracking in pre-stressed concrete beam? | 2M |
| (i) What will happen to a cantilever beam, if it is supported by a column at free end after construction? | 2M |
| (j) What is composite beam? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain with neat sketch Hoyer system of pre tensioning. 10M
- OR**
- 3 What are different post tensioning systems? Explain in detail. 10M
- 4 A post tensioned cable of a beam 10 m long is initially tensioned to as stress of 800 N/mm^2 at one end. If the tendons are curved so that the slope is 1 in 30 at each end, with cross sectional area of 600 mm^2 . Calculate the loss of pre-stress due to friction. Take coefficient of friction between duct and cable as 0.25 and due to wave effect as 0.0012 per metre. 10M
- OR**
- 5 A rectangular beam of 12 m long and cross section 350 mm x 600 mm is post-tensioned by straight tendons tensioned to a stress of 900 N/mm^2 at the jacking end. Calculate the wobble friction loss at the mid span and at the end remote from the jack. Take Wobble coefficient = 0.25 per 100 metre. 10M
- 6 A rectangular beam 100 mm wide x 250 mm deep spanning over 8 m is pre-stressed by a straight cable carrying an effective pre-stressing force of 250 kN located at an eccentricity of 40mm. The beam supports live load of 1.2 kN/m . Take unit weight of concrete 25 kN/m^3 . 10M
- (i) Calculate the resultant stress distribution for the central cross section of the beam,
- (ii) Find the magnitude of the pre-stressing force with an eccentricity of 40 mm which can balance the stress due to dead load and live load at bottom fibre of the central section of the beam.

OR**Contd. in Page 2**

- 7 A pre-stressed beam with width 150 mm and depth 350 mm, having span 6 m is subjected to uniformly distributed load of 4 kN/m including self weight of beam. Beam is pre-stressed by a cable carrying a force 200 kN and at an eccentricity of 50 mm. Determine the location of thrust line giving position at quarter and central span sections. 10M
- 8 Why deflection in beam is to be controlled? 10M
- OR**
- 9 What are the factors affecting short term deflection in pre-stressed concrete beam? 10M
- 10 State any four IS requirements related to pre-stressed concrete composite beams. 10M
- OR**
- 11 Explain the shrinkage stresses developed in composite beam and slab construction. 10M

B.Tech IV Year I Semester (R19) Supplementary Examinations December 2024

PRESTRESSED CONCRETE

(Civil Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|--|----|
| (a) Explain the principle of pre-stressing and its significance in construction. | 2M |
| (b) What are the advantages of using pre-stressed concrete over conventional concrete? | 2M |
| (c) Define pre-tensioning and describe how it differs from post-tensioning. | 2M |
| (d) What is the Hoyer system, and how is it used in pre-tensioning? | 2M |
| (e) Discuss the limitations of pre-stressed concrete. | 2M |
| (f) What factors contribute to the loss of pre-stress in pre-tensioned members? | 2M |
| (g) Explain the concept of the Kern line in the analysis of pre-stressed beams. | 2M |
| (h) Describe the differences between straight and parabolic tendons in pre-stressing. | 2M |
| (i) What are the primary causes of long-term deflections in pre-stressed concrete beams? | 2M |
| (j) Define composite beams and explain the concept of differential shrinkage. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | | |
|-----------|---|----|
| 2 | (a) Discuss the limitations of pre-stressed concrete. | 5M |
| | (b) Explain the Freyssinet and Gifford-Udall systems of post-tensioning. | 5M |
| OR | | |
| 3 | (a) What materials are required for high-strength pre-stressed concrete? | 5M |
| | (b) Explain the need for pre-stressing in modern construction. | 5M |
| 4 | (a) What are the causes of relaxation in pre-stressing steel? | 5M |
| | (b) Explain the impact of anchorage slip and friction on pre-stress loss. | 5M |
| OR | | |
| 5 | (a) Describe the methods to measure and calculate shrinkage losses in concrete. | 5M |
| | (b) A post-tensioned beam experiences a total loss of pre-stress due to relaxation of 20% of the initial pre-stressing force. If the initial pre-stress was 1000 kN, calculate the final pre-stressing force. | 5M |
| 6 | (a) Analyze a beam pre-stressed with parabolic tendons for flexure. | 5M |
| | (b) Explain the design process for an I-section pre-stressed concrete beam as per IS 1343. | 5M |
| OR | | |
| 7 | (a) Analyze a beam pre-stressed with straight tendons for shear. | 5M |
| | (b) Discuss the significance of transmission length in pre-stressed members. | 5M |
| 8 | (a) Describe the factors influencing deflections in pre-stressed members. | 5M |
| | (b) Discuss the methods to control long-term deflections in uncracked beams. | 5M |
| OR | | |
| 9 | (a) A pre-stressed beam of span 5 m is subjected to a uniformly distributed load of 15 kN/m. If the moment of inertia is $150 \times 10^6 \text{ mm}^2$, calculate the mid-span deflection. | 5M |
| | (b) Explain the factors that influence the deflection of pre-stressed concrete members. | 5M |
| 10 | (a) Explain the stress distribution in propped and un-propped composite beams. | 5M |
| | (b) What are the effects of differential shrinkage on composite beams? | 5M |
| OR | | |
| 11 | (a) Discuss the analysis of stress in composite beams under differential shrinkage. | 5M |
| | (b) What are the challenges in designing propped composite beams? | 5M |

B.Tech IV Year II Semester (R15) Supplementary Examinations December 2024

PRESTRESSED CONCRETE

(Civil Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- (a) What is the grade of concrete used for post tensioned concrete members?
 - (b) State one distinctive feature of the Gifford-Udall System in prestressed concrete.
 - (c) List the types of losses in prestressed concrete beam.
 - (d) How do you predict the loss of prestress in PSC beams?
 - (e) What is cracking moment? On what factors does it depend?
 - (f) List the types of losses in prestressed concrete beam.
 - (g) How do you improve shear resistance by pre-stressing techniques?
 - (h) Explain composite section.
 - (i) State any two factors influencing the deflection.
 - (j) What is short term deflection?

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Describe the key milestones in the historic development of prestressed concrete.
(b) How did the introduction of prestressed concrete impact the construction industry?

OR

- 3 (a) Enumerate the advantages of using prestressed concrete in structural engineering.
(b) Highlight the limitations or challenges associated with prestressed concrete.

- 4 (a) Discuss about the loss of pre -stress due to bending of the member.
(b) A post tensioned PSC beam of 100 mm x 300 mm size is stressed by a parabolic cable with zero eccentricity at the supports and 50 mm at centre of the span. The area of the cable is 200 mm² and initial stress in the cable is 1200 N/mm². If the modulus of elasticity of steel is 210 kN/mm² and ultimate creep strain is 30×10^{-6} per N/mm² of stress, compute the loss of stress in steel only due to creep of concrete.

OR

- 5 A pre -tensioned PSC beam of 250 mm x 300 mm size is stressed by 12 wires of 7 mm diameter initially stressed to 1200 N/mm² with their centroids located 100 mm from soffit. Estimate the final percentage loss of stress due to elastic deformation, creep, shrinkage and relaxation using IS-1343 and following data:

Relaxation of steel stress = 90 N/mm²; $E_s = 210 \text{ kN/mm}^2$; $E_c = 35 \text{ kN/mm}^2$;

Creep coefficient = 1.6 and Residual shrinkage strain = 3×10^{-4} .

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- 6 (a) Explain the elastic design of PSC beams of rectangular and I-section for flexure.
(b) A pre-stressed beam has a symmetrical I - section in which the depth of each flange is one - fifth of overall depth and the web is thin enough to be neglected in bending calculations. At the point of maximum bending moment, the pre - stressing force is located at the centre of the bottom flange and the total loss of pre -stress is 20%. If there is to be no tensile stress in the concrete at any time, show that the dead load must be at least one -seventh of live load.

OR

- 7 An unsymmetrical I-section beam is used to support an imposed load of 2 kN/m over a span of 8 m. The sectional details are top flange 300 mm wide and 60 mm thick; overall depth of the beam = 400 mm. At the centre of the span, the effective prestressing force of 100 kN is located at 50 mm from the soffit of the beam. Estimate the stresses at the centre of span section of the beam for the following load conditions.
(i) Prestress + self – weight,
(ii) Prestress + self - weight + live load.

- 8 (a) How will you improve the shear resistance of structural concrete members by applying prestressing technique?
(b) Explain the design of shear reinforcement in the context of IS - 1343 code provisions.

OR

- 9 (a) Discuss in detail about the general design considerations in respect of composite section of PSC.
(b) A PSC beam of 6 m span and 150 mm x 300 mm in size is pre - stressed by a straight cable carrying an effective force of 220 kN at an eccentricity of 50 mm. If the modulus of elasticity of concrete is 38 kN/mm^2 , compute the deflection at centre of the span due to pre - stress and self - weight and find the magnitude of uniformly distributed live load which will nullify deflection due to pre -stress and self-weight.
- 10 (a) Discuss IS code requirement for maximum deflections of PSC members.
(b) A PSC beam of 150 mm x 300 mm size is simply supported over a span of 8 m and pre - stressed by a symmetric parabolic cable at a distance of 75 mm from the bottom of beam at mid - span and 125 mm from top of the beam at supports. If $E_c = 38 \text{ kN/mm}^2$ and force in the cable is 350 kN, compute the deflection at mid -span when the beam is supporting its own weight.

OR

- 11 List out the factors and explain them in detailed different parameters that influence the long term and short-term deflections of a PSC member subjected to flexure.

B.Tech IV Year II Semester (R13) Supplementary Examinations August 2022

PRESTRESSED CONCRETE

(Civil Engineering)

(For 2014 regular admitted batches only)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- What are the advantages of prestressed concrete?
 - List out the various systems of prestressing.
 - What do you mean by loss of prestress due to elastic shortening of concrete?
 - What are the different losses of prestress present in pre-tensioned and post-tensioned members?
 - What is the importance of cable profile in prestressed members?
 - Define kern line with neat sketch.
 - What is differential shrinkage?
 - Define principal stress.
 - What are the factors influencing the long term deflections of prestressed concrete members?
 - List out the different methods to control deflection of prestressed concrete members.

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Explain Hoyer system of prestressing with the help of neat sketches.
(b) Discuss in detail about the pretensioning and post tensioning methods.

OR

- 3 (a) Discuss Magnel system of prestressing with the help of neat sketches.
(b) Explain the historical development of pre stressed concrete.
- 4 (a) Explain in detail about slip in anchorage and wobble frictional losses.
(b) A prestressed concrete pile is 300 mm × 300 mm in section and is provided with 40 wires of 3 mm diameter distributed uniformly over the section. Initially the wires are tensioned in prestressing beds with a total pull of 450 kN. Determine the final stress in concrete and the percentage loss of stress in the wires.

Take:

$$E_s = 2.08 \times 10^5 \text{ N/mm}^2$$

$$E_c = 3.2 \times 10^4 \text{ N/mm}^2$$

$$\text{Creep shortening} = 32 \times 10^{-6} \text{ mm/mm per N/mm}^2 \text{ of stress.}$$

$$\text{Total shrinkage strain} = 200 \times 10^{-6}.$$

$$\text{Relaxation loss of stress in steel} = 4.5 \text{ percent of the initial stress.}$$

OR

- 5 (a) Discuss in detail about losses of prestress due to shrinkage and creep of concrete.
(b) Explain various failure modes of prestressed concrete beam when subjected to bending.

Contd. in page 2

- 6 (a) A post tensioned bridge girder with unbounded tendons is of size 1200 mm wide by 1800 mm deep is of box section with wall thickness of 150 mm. The high tensile steel has an area of 4000 mm^2 and is located at an effective depth of 1600 mm. The effective prestress in steel after loss is 1000 N/mm^2 & effective span is 24 m. If $f_{ck} = 40 \text{ N/mm}^2$, $f_y = 1600 \text{ N/mm}^2$. Estimate the flexural strength.
- (b) Explain the design procedure of I-section for flexure according to IS code.

OR

- 7 (a) A pretensioned prestressed rectangular concrete beam of 200 mm wide and 400 mm deep has an effective cover of 55 mm. If $f_{ck} = 30 \text{ N/mm}^2$ and $f_p = 1700 \text{ N/mm}^2$ and the area of prestressing steel $A_{st} = 520 \text{ mm}^2$, calculate the ultimate flexural strength of the section using codal provisions.
- (b) Explain in detail about various types of tendons.
- 8 (a) A PSC beam of rectangular section 300 mm wide and 600 mm deep is pre stressed by a parabolic cable located at an eccentricity of 120 mm at mid span and zero at the supports. If the beam has a span of 10 m and carries a uniformly distributed live load of 8 kN/m, design the shear reinforcement at the supports and quarter span section. Grade of concrete is M40. Yield stress of tendons is 2000 MPa. Area of the tendons is 1000 mm^2 . Effective stress in the tendons is 1500 MPa.
- (b) Explain in detail about the design procedure for shear in beams.

OR

- 9 (a) Explain the importance of differential shrinkage in composite sections.
- (b) The support section of a prestressed concrete beam, 100 mm wide by 250 mm deep, is required to support an ultimate shear force of 80 kN. The compressive prestress at the centroidal axis is 5 N/mm^2 . The characteristic cube strength of concrete is 40 N/mm^2 . The cover to the tension reinforcement is 50 mm. If the characteristic tensile strength of stirrups is 415 N/mm^2 , design suitable shear reinforcements in the section using IS code recommendations.
- 10 (a) A concrete beam having a rectangular section $100 \times 300 \text{ mm}$ is prestressed by a parabolic cable with an initial prestressing force of 240 kN. The cable has an eccentricity of 50 mm at the centre and concentric at the supports. If the span of the beam is 12 m and subjected to a live load of 5 kN/m. Calculate the short term deflection at midspan. Assume $E_c = 38 \text{ kN/mm}^2$, Creep coefficient = 2, Loss of prestress = 20%. Estimate the long-term deflection.
- (b) Explain in detail about the factors influencing short term deflections of prestressed concrete members.

OR

- 11 (a) Discuss in detail about importance of control of deflection in prestressed concrete beams.
- (b) A PSC beam of rectangular section 120 mm wide and 300 mm deep spans over 6 m. The beam is prestressed by a straight cable carrying an effective force of 180 kN at an eccentricity of 50 mm. If it supports an imposed load of 4 kN/m and modulus of elasticity of concrete is 38 kN/mm^2 , compute the upward deflection under prestress and self weight.
